

① First Class:

-Def. A polytope, P , is the convex hull of a set of vertices in $\mathbb{R}^N$. i.e. $v_1, \dots, v_k \in \mathbb{R}^N \Rightarrow \left\{ \sum_{i=1}^k \lambda_i v_i \mid 0 \leq \lambda_i \leq 1 \right\}$

-Exs:



"Simplicial": all proper facets are simplexes Δ , $\triangle$, etc.

Face vector: $f = (f_0, f_1, \dots, f_{d-1})$, $f_i = \#$ of i -dimensional faces

Q: Given a vector f , when is there an associated polytope?

[Simpler: can we answer for what f there is an associated simplicial polytope?]

McMullen: Better to associate an h -vector to P .

→ Inductive definition (induction on faces of P)

→ For simplicial P , $h(P) = (h_0, \dots, h_d)$ are coeffs. of

$$h(P, t) = (t-1)^d + f_0(t-1)^{d-1} + \dots + f_{d-1}$$

e.g. $P = \Delta_n$ a n -diml simplex

$$\Rightarrow h(\Delta_n, t) = t^d + t^{d-1} + \dots + 1$$

$P = \text{cube}$

$$\Rightarrow h(P, t) = t^3 + 5t^2 + 5t + 1$$

Conjecture (McMullen) / Thm (Stanley): $h \in \mathbb{Z}_{\geq 0}^{d+1}$ is an h -vector for a d -diml

simplicial polytope P iff

- ① $h_i = h_{d-i} \quad \forall i$
- ② $\exists$ graded $\mathbb{Q}$ -alg. R with $R_0 = \mathbb{Q}$, R generated in deg 1,
 $\dim_{\mathbb{Q}} R_i = h_i - h_{i-1} \quad \forall 1 \leq i \leq \lfloor d/2 \rfloor$

Cor. P : simplicial $\Rightarrow h(P)$ is unimodal

Stanley's idea: $P \longleftrightarrow \text{"fans"} \longleftrightarrow \text{toric variety } X_P / \mathbb{C}$

Thm: P : simplicial $\Rightarrow h(P, t) = \sum_{k=0}^d \dim H^{2k}(X_P, \mathbb{Q}) \cdot t^k$
 $\& H^{\text{odd}}(X_P, \mathbb{Q}) = 0$

When P is not simplicial, X_P is not " $\mathbb{Q}$ -smooth" & Stanley's theorem fails. But there is a fix!

② Thm (Stanley): For general rational $\mathbb{P}$,

$$h(\mathbb{P}, t) = \sum_{k=0}^d \underbrace{\dim \mathrm{IH}^{2k}(X_{\mathbb{P}}, \mathbb{Q})}_{\text{"Intersection homology"}} \cdot t^k$$

~ We will return later in the course to prove this. ~

What is "Intersection (co)homology"?

A: provides a good (co)homology theory for singular varieties
 "good": we can do intersection theory with it.

Intersection Theory for smooth varieties:

X : smooth, oriented n -manifold, $H_n(X; \mathbb{Z}) \cong \mathbb{Z} \ni [X]$ fundamental class

Thm (Poincaré Duality). $\cap [X]: H^i(X, \mathbb{Z}) \rightarrow H_{n-i}(X, \mathbb{Z})$

So with rational coeffs:

$$\begin{array}{ccc} H^i(X, \mathbb{Q}) & \xrightarrow{\sim} & H_{n-i}(X, \mathbb{Q}) \\ \parallel \cong & \text{universal coeffs. thm} & \parallel \cong \\ H_i(X, \mathbb{Q})^* & & H^{n-i}(X, \mathbb{Q})^* \end{array}$$

Cor: $h_i = \dim_{\mathbb{Q}} H_i(X, \mathbb{Q}) = \dim_{\mathbb{Q}} H^{n-i}(X, \mathbb{Q})$ is a symmetric/palindromic sequence.

Better: can specify the isomorphism $H^{n-i}(X, \mathbb{Q}) \cong H^{n+i}(X, \mathbb{Q})$

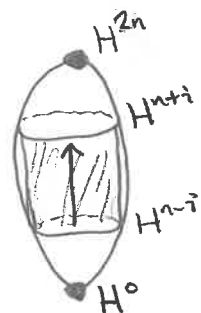
Assume $X \subseteq \mathbb{P}_{\mathbb{C}}^N$ is a complex, projective variety.

Choose a generic hyperplane $H \subset \mathbb{P}^N$

$$\leadsto [X \cap H] \in H_{2n-2}(X, \mathbb{Z}) \xLeftrightarrow{\text{P.D.}} [H] \in H^2(X, \mathbb{Z})$$

Thm (Hard Lefschetz). $H^{n-i}(X, \mathbb{Q}) \xrightarrow[\sim]{- \cup [H]^i} H^{n+i}(X, \mathbb{Q})$ is isom.

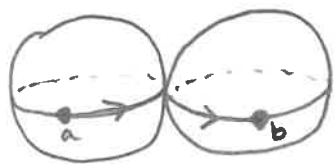
↳ Rmk on how "hard" Hard Lefschetz is: hard enough Lefschetz didn't prove it in his 1924 paper; Hodge did in 1941!



Cor: h_i are also unimodal.

③ This fails for singular spaces

Ex: $X = \mathbb{P}^1 \cup \mathbb{P}^1 \subseteq \mathbb{P}^2$



By Mayer-Vietoris:

$$H_i(X, \mathbb{Q}) = \begin{cases} \mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q} \oplus \mathbb{Q} & i=2 \end{cases} \quad \text{no Poincaré duality}$$

$H^i(X, \mathbb{Q})^*$

What is going on? a & b want to be distinct 0-cycles, but they are connected by a 1-cocycle through the singularity.

IH will exclude this cocycle:

$$IH^i(X, \mathbb{Q})^* \cong IH_i(X, \mathbb{Q}) = \begin{cases} \mathbb{Q} \oplus \mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q} \oplus \mathbb{Q} & i=2 \end{cases}$$

But what if I only care about cohomology? IH has a sheafy analogue called perverse sheaves.

cohomology sheaf of a proper map $\in \left\{ \begin{array}{l} \text{perverse} \\ \text{sheaves} \end{array} \right\}$

The category of perverse sheaves has atoms called

"Intersection Cohomology" (IC) sheaves.

Thm [Decomposition Thm]. $f: X \rightarrow B$ a proper map between smooth varieties, then $Rf_* \mathbb{Q}_X$ is a semisimple perverse sheaf, i.e. it is a direct sum of IC sheaves.

$\Rightarrow$ If you know the supports of summands, you can go from general fibers to smooth fibers!

Underlies Springer theory, proof of Fundamental Lemma, Kazhdan-Lusztig theory, proof of P=W conjecture, ... many many more!

(3.5)

Recap of some algebraic topology:Chains: $\Delta^n = n$ -simplex $C_n(X) =$ free abelian group generated by maps $\Delta^n \rightarrow X$

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \cdot (\sigma \circ d_i) \quad \text{for } d_i: \Delta^{n-1} \hookrightarrow \Delta^n \text{ the } i\text{th face.}$$

$$H_i(X) = H_i(\dots \rightarrow C_n(X) \rightarrow C_{n-1}(X) \rightarrow \dots)$$

Rmk: Can replace $C_n(X)$ with set of (possibly infinite) sums $\sum c_i \sigma_i$ s.t. every cpt $K \subset X$ intersects only finitely many σ_i . In this case, the complex computes Borel-Moore homology, a sort of "homology w/cpt supports".

e.g. $\sigma_i: \Delta^1 \rightarrow \mathbb{R}$ by the interval $[i, i+1]$

$$\sum_{i \in \mathbb{Z}} \sigma_i \in C_1^{BM}(\mathbb{R})$$

Exercise: $\sum_{i \in \mathbb{Z}} \sigma_i$ generates $H_1^{BM}(\mathbb{R}) \cong \mathbb{Z}$.

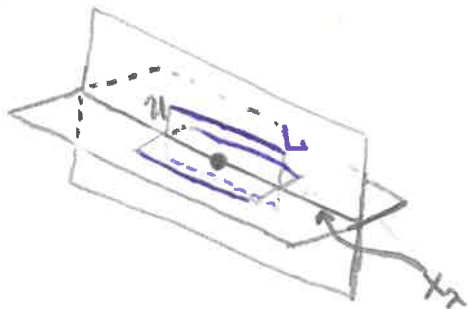
Whitney Stratifications (CORRECTED): $n = \dim_{\mathbb{R}}(X)$

$X = \bigsqcup_{\lambda} X_{\lambda}$ is a stratification if $\overline{X_{\lambda}}$ is a union of X_{μ} for some set of μ .

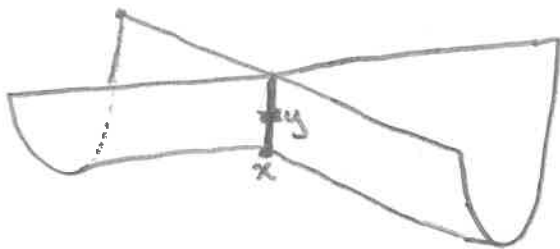
This stratification is a Whitney stratification if

$\forall \lambda$, there is a stratified space $L = \bigsqcup_{\mu} L_{\mu}$ of $\dim(L) = n - \dim(X_{\lambda}) - 1$ s.t. $\forall x \in X_{\lambda}$, $\exists$ neighborhood $\mathcal{U} \ni x$ s.t. $\phi: \mathcal{U} \xrightarrow{\text{homeo.}} \mathbb{R}^{\dim(X_{\lambda})} \times \text{cone}(L)$, where
 $\text{cone}(L) = L \times [0, 1] / L \times \{0\}$

$\& \phi^{-1}(L_{\mu}) = X_{\mu} \cap \mathcal{U} \& \phi|_{\mathcal{U} \cap X_{\lambda}}: \mathcal{U} \cap X_{\lambda} \xrightarrow{\sim} \mathbb{R}^{\dim(X_{\lambda})} \times [0, 1]$



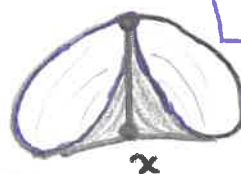
Ex: Whitney Umbrella: $\{x^2 = zy^2\} \subseteq \mathbb{C}^3$



At y:

$y \in \mathcal{U}: \mathcal{U} \cap X_{\lambda} \approx \mathbb{R} \times \text{cone}(L)$

At x:



$L = S^1 \vee S^1 = (S^1 \vee S^1 \setminus \{\text{pt}\}) \sqcup \{\text{pt}\}$
 (for real picture)

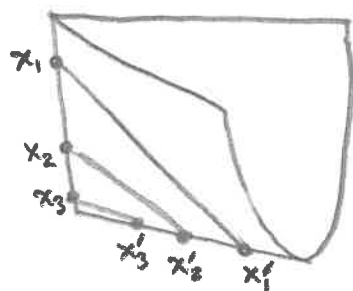
These are equivalent to the Whitney conditions:

(a) If $x_i \rightarrow x$ for $x_i \in X_2$ & $x \in X_M$, then

$$T_x X_M \subseteq \lim_{i \rightarrow \infty} T_{x_i} X_2 \text{ if this limit exists}$$

(b) If $x_i \in X_2$ & $x'_i \in X_M$ both go to $x \in X_2$, then $\lim_{i \rightarrow \infty} [\text{Secant line } (x_i, x'_i)] \subseteq \lim_{i \rightarrow \infty} T_{x_i} X_2$ if limits exist.

Ex: for Whitney umbrella, can take



} limit of secant lines is horizontal, but $\lim_{i \rightarrow \infty} T_{x_i} X_2$ is vertical line only

So we need extra strata.

(Complex) Link of Whitney Umbrellas:

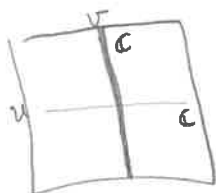
$$\pi: \mathbb{C}^2 \rightarrow X$$

$$(u,v) \mapsto (uv, v, u^2)$$

compatible w/ contraction

$$\lambda \cdot (u,v) = (\lambda u, \lambda^2 v) \quad \& \quad \lambda \cdot (x,y,z)$$

$$= (\lambda^3 x, \lambda^2 y, \lambda^2 z)$$



glues $\pm u$ coords.

$$\text{Link at } (0,0) \text{ is } X \setminus \{0\} / G_m \cong \mathbb{C}^2 \setminus \{0\} / \left(\begin{array}{l} (u,0) = (-u,0) \forall u \\ \lambda(u,v) = (u,v) \forall \lambda \neq 0, v \end{array} \right)$$

$$\cong S^3 / \left(\begin{array}{l} (u,0) = (-u,0) \\ \text{on an unknotted} \\ S^1 \subset S^3 \end{array} \right)$$

Links of Isolated Singularities:

Thm (Milnor fibers). $f: \mathbb{C}^{n+1} \rightarrow \mathbb{C}$ s.t. $f(0) = 0$, f : holomorphic

Let $B_\varepsilon = \{z \in \mathbb{C}^{n+1} \mid \|z\| \leq \varepsilon\}$ be a small ball @ $0 \in \mathbb{C}^{n+1}$

Then $f^{-1}(0) \cap B_\varepsilon \underset{\text{homeo.}}{\approx} \text{cone}(L)$ for $L \underset{\mu}{\approx} \bigvee S^n$.
(homotopic)

$$\text{for } \mu = \dim_{\mathbb{C}} \frac{\mathbb{C}[x_0, \dots, x_{n+1}]}{(\frac{\partial f}{\partial x_0}, \dots, \frac{\partial f}{\partial x_{n+1}})}$$

Ex: $f: \mathbb{C}^2 \rightarrow \mathbb{C}$ has $\mu = \dim_{\mathbb{C}} \frac{\mathbb{C}[x,y]}{(2y, -3x^2)} = \dim_{\mathbb{C}} S_{\text{ann}\{1,x\}} = 2$
 $(x,y) \mapsto y^2 - x^3$

$$L \approx S^1 \vee S^1$$

Complex Link of $X = \{y^2 = x^3\} \subseteq \mathbb{C}^2$:

$$\pi: \mathbb{C} \rightarrow X$$

$$z \mapsto (z^2, z^3)$$

Write $z = re^{i\theta}$ & fix r .

$\Rightarrow (r^2 e^{2i\theta}, r^3 e^{3i\theta})$ traverses 1 circle 3 x's & the other twice.

$\Rightarrow$ torus knot in S^3 .

④

LECTURE 2:

Intersection Homology: X : singular $\mathbb{C}$ -variety.

0: Sing. are interesting

e.g. $x^n = y^m$

1: Whitney Stratifications

$X = \coprod_{\lambda} X_{\lambda}$ stratification (ie. X_{λ} conn'd, locally clsd, smooth)
 $\overline{X_{\lambda}} = \bigcup_{\mu} X_{\mu}$

We ask X_{λ} is "equisingular", i.e. $\forall x \in X_{\lambda}, \exists$ ngbd $U \ni x$ s.t. $(X_{\lambda} \cap U) \times V \cong U$ (id on $X_{\lambda} \cap U$) ("local normal triviality")

Ex: Whitney umbrella $\{x^2 = zy^2\} \subseteq \mathbb{C}^3$.



$(u,v) \mapsto (uv, v, u^2)$

$X_{reg} \cup X_{sing}$ not good enough!

$X_{reg} \cup (X_{sing} \setminus \{0\}) \cup \{0\}$

$y^2 + x^3 - t^2 x^2 = 0$:



Whitney Condition: $X_{\mu} \subset \overline{X_{\lambda}} \Rightarrow \forall a_i \in X_{\lambda}, b_i \in X_{\mu}$

converging to $c \in X_{\mu} \Rightarrow \lim_{i \rightarrow \infty} (\secants a_i - b_i) \in \lim_{i \rightarrow \infty} T_{b_i} X$

Intersection Homology: to recover P.D. / intersection theory need to restrict what cycles we deal with.

Def: A perversity is a fn $\overline{p}: [0, \dim X] \cap \mathbb{Z} \rightarrow \mathbb{N}$ s.t. $p_{\alpha} := \overline{p}(\alpha)$ satisfies $\begin{cases} p_0 = p_1 = p_2 = 0 \\ p_{\alpha} \leq p_{\alpha+1} \leq p_{\alpha} + 1, \alpha \geq 2 \end{cases}$

real dimension



slopes 0 or 1 everywhere.

e.g. upper middle perversity: $(0, 0, 0, 1, 1, 2, 2, \dots, \frac{n}{2}-1)$

lower middle: $(0, 0, 0, 0, 1, 1, 2, 2, \dots, \frac{n}{2}-1, \frac{n}{2}-1)$

$X = X_n \supset X_{n-1} \supset \dots \supset X_1 \supset X_0 \supset X_{-1} = \emptyset$ ($\dim X_i \leq i$)

An i -chain $\xi \in C_i(X)$ is $\overline{p}$ -allowed if $\dim(\xi \cap X_{n-\alpha}) \leq i - \alpha + p_{\alpha}$

Note: $\xi \not\cap X_{n-\alpha} \iff \dim(\xi \cap X_{n-\alpha}) = i - \alpha$

$IC_{\overline{p}}^i(X) = \{ \xi \in C_i(X) \mid \begin{cases} \dim(\xi \cap X_{n-\alpha}) \leq i - \alpha + p_{\alpha} \\ \dim(\xi \cap X_{n-\alpha}) \leq i - 1 - \alpha + p_{\alpha} \end{cases} \forall \alpha \geq 2 \}$

$IH_{\overline{p}}^i(X; \mathbb{Z}) = H_i(IC_{\overline{p}}^{\bullet}(X))$

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e.g.



don't want this one!

★ DEFAULT

⑤. Ex: $X = P^1 \cup P^1 \subseteq P^2$, $X_2 = A^1 \cup A^1 \supseteq X_1 = X_0 = \{pt\}$

$P_0 = P_1 = P_2 = 0$ is only option for perversity fn.

$$IC_i(X) = \left\{ \xi \in C_i(X) \mid \begin{array}{l} \dim(|\xi| \cap X_0) \leq i-2 \\ \dim(|\partial\xi| \cap X_0) \leq i-3 \end{array} \right\}$$

Chains can only touch singularity at $i=2$ (impossible due to this)

$$\Rightarrow IC_*(X) \cong C_*(P^1 \sqcup P^1)$$

$$\text{So } IH_i(X) = H_i(P^1 \sqcup P^1) = \begin{cases} \mathbb{Q} \oplus \mathbb{Q} & i=0,2 \\ 0 & i=1 \end{cases}$$

More generally:

Ex: X is $2m$ -dimensional with an isolated singular point x
 $X = X_{2m} \supset X_0 = \{x\}$.

Let's use the middle perversity

Prop: $IH_i(X, \mathbb{Z}) = \begin{cases} H_i(X, \mathbb{Z}), & i > m \\ \text{Im}(H_m(X-x, \mathbb{Z}) \rightarrow H_m(X, \mathbb{Z})), & i = m \\ H_i(X-x, \mathbb{Z}), & i < m \end{cases}$

Pf: $\xi \in IC_i^m(X)$ iff ① $\dim(|\xi| \cap X_0) \leq i - 2m - \underline{m}(2m) = i - m - 1$
 ② $\dim(|\partial\xi| \cap X_0) \leq i - m - 2$

If $i \leq m$, can't intersect $X_0 = \{x\} \Rightarrow IC_i^m(X) = C_i(X-x)$

If $i \geq m+2$, no condition $\Rightarrow IC_i^m(X) = C_i(X)$

If $i = m+1$, ξ can go through x , but not $\partial\xi$

$$IC_*(X) : \dots \rightarrow C_{m+3}(X) \rightarrow C_{m+2}(X) \rightarrow IC_{m+1}(X) \rightarrow C_m(X-x) \rightarrow \dots$$

$$\quad \quad \quad \hookrightarrow C_{m-1}(X-x) \rightarrow \dots$$

Result is clear for $i \neq m, m+1$.

$$\text{Ker} \left[\underset{\substack{\uparrow \\ C_{m+1}(X)}}{IC_{m+1}(X)} \rightarrow C_m(X-x) \right] = \text{Ker} [C_{m+1}(X) \rightarrow C_m(X-x)]$$

since if $\partial\xi = 0$, $x \in |\partial\xi|$

$\Rightarrow$ result holds at $m+1$

$$\partial : IC_{m+1}(X) \rightarrow \partial C_{m+1}(X) \cap \underbrace{IC_m(X)}_{C_m(X-x)}$$

$$\Rightarrow IH_m \text{ is image of } H_m(X-x) \text{ in } H_m(X)$$

⑥ Cor. $X = \text{pinched torus} = \{y^2 = x^3 - x^2\} \subseteq \mathbb{P}^2$ has

$$IH_i(X, \mathbb{Z}) = \begin{cases} \mathbb{Z} & i=0,2 \\ 0 & i=1 \end{cases}$$

Pf. $H_1(\underbrace{X-x}_{\cong S^1}, \mathbb{Z}) \cong \mathbb{Z} \longrightarrow H_1(X, \mathbb{Z})$



$$\begin{array}{ccc} 1 & \xrightarrow{\quad} & 0 \\ & \text{bc gen. is} & \\ & \text{a bdy in } X & \end{array}$$

//

~~Ex: (Why we need a persistence function at all)~~ ← Will return to this later

- Computing with chains is hard; often computed via normalizations.

Normalization: "a type of partial compactification"

Def. A domain A is normal if it is integrally closed in its field of fractions.

ie. $F = \text{Frac}(A) \supseteq A$. A is normal if for every $a_1, \dots, a_m \in A$ & every $x \in F$ s.t. $x^m + a_1 x^{m-1} + \dots + a_m = 0$ we have $x \in A$.

Def. The normalization of A is

$\{x \in F \mid x \text{ satisfies } x^m + a_1 x^{m-1} + \dots + a_m = 0 \text{ for some } m \text{ & } a_i \in A\}$

⑦ Ex: $\{x^2 = y^3\} \subseteq \mathbb{C}^2$ (has singularity at $(0,0)$)

$$\Downarrow$$

$$A = \mathbb{C}[x,y]/(x^2 - y^3) \cong \mathbb{C}[t^3, t^2] \subseteq \mathbb{C}[t]$$

$$\text{Frac}(A) \cong \mathbb{C}(t)$$

t satisfies $t^3 - (x) = 0$ & $\bar{A} = \mathbb{C}[t]$

$$A \longrightarrow \bar{A} \iff \{x^2 = y^3\} \longleftarrow \mathbb{A}^1$$

Normality is related to resolutions of singularities (not equiv. to)

Prop: If A is ^(noetherian) normal, then A is "regular in codim 1"
 i.e. for every height 1 prime $\mathfrak{p} \subseteq A$, $A_{\mathfrak{p}}$ is regular.
 (Geometrically, the singular locus of A is $\text{codim} \geq 2$.)

Pf: Since $A_{\mathfrak{p}}$ is normal, it is sufficient to show?

Claim: If A is a normal, noetherian local ring of dimension 1, then its maximal ideal $\mathfrak{m} \subseteq A$ is principal.

Pf of Claim: $x \in \mathfrak{m}$ nonzero $\Rightarrow \mathfrak{m} = \sqrt{(x)} \Rightarrow \mathfrak{m}^n \subseteq (x)$ some n

Choose n smallest s.t. $\mathfrak{m}^n \subseteq (x)$, so $\exists y \in \mathfrak{m}^{n-1}$ s.t. $y \notin (x)$
 $\Rightarrow t = \frac{y}{x} \in \text{Frac}(A) \setminus A$ & $t \cdot \mathfrak{m} \subseteq A$ so $t\mathfrak{m} = A$ or $\subseteq \mathfrak{m}$.

If $t\mathfrak{m} \subseteq \mathfrak{m} \Rightarrow t\mathfrak{m}$ is f.g. A -module & is $A[t]$ -module

$\Rightarrow t$ is integral $/A \Rightarrow$ contradiction.

So $t\mathfrak{m} = A$ & $\exists m \in \mathfrak{m}$ s.t. $tm = 1 \Rightarrow \mathfrak{m} = A \cdot t^{-1} = A \cdot m$
 is principal //

So the normalization is a "partial resolution".

What does normality look like?

Def. X is topologically normal if $\forall x \in X$, $\exists$ open nbd $U_x \ni x$
 s.t. $U_x - X_{n-2} = U_x - \text{Sing}(X)$ is connected.

Ex:  is not topologically normal, but

 is.

⑦ Prop. X normal $\Rightarrow X$ topologically normal.

Pf: We use a key property of normal varieties:

Zariski's Main Thm: If $f: Z \rightarrow X$ is a proper, birat'l map, X is normal & Z is irreducible, then ① $f^{-1}(x)$ is connected $\forall x \in X$, ② $f_* \mathcal{O}_Z = \mathcal{O}_X$.

Hironaka's Thm: Any quasiprojective variety X admits a (proper) resolution of singularities $f: Z \rightarrow X$.

We argue by passing to $Z: x \in U$ ^{analytic} nhd of x . Then put $E = \pi^{-1}(U) \cap \pi^{-1}(\text{Sing}(X))$. E is clsd in Z & Z is smooth, so $\pi^{-1}(U) \setminus E$ is still connected if $\pi^{-1}(U)$ is connected (in analytic top) (bc $\pi^{-1}(U) = V_1 \cup V_2 \Rightarrow \pi(V_1) \cap \pi(V_2) = \emptyset \Rightarrow U = \pi(V_1) \cup \pi(V_2)$)
 $U \setminus \pi(U)$ — because f is a resolution of singularities //

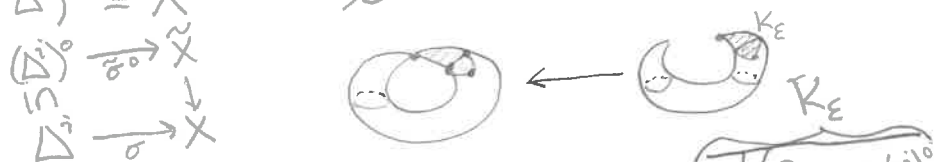
Thm (Goresky-MacPherson) $\pi: \tilde{X} \rightarrow X$: the normalization of X . Then $\pi_*: IH_i^{\bar{P}}(\tilde{X}; \mathbb{Z}) \rightarrow IH_i^{\bar{P}}(X; \mathbb{Z})$ is an isom.

Pf: π induces a map on complexes $\pi_*: IC_i^{\bar{P}}(\tilde{X}, \mathbb{Z}) \rightarrow IC_i^{\bar{P}}(X, \mathbb{Z})$ which takes a chain $\sigma: \Delta^i \rightarrow \tilde{X}$ to $\pi \sigma$.

If $X_n \supseteq X_{n-1} \supseteq \dots \supseteq X_0$ is strat of X , we can take preims. to get $\tilde{X}_n \supseteq \tilde{X}_{n-1} \supseteq \dots \supseteq \tilde{X}_0$, which suffices to compute $IC_i^{\bar{P}}(\tilde{X})$ (Exercise: $\tilde{X}_n \supseteq \dots \supseteq \tilde{X}_0$ is a Whitney stratification) See Back

We claim any chain $\sigma: \Delta^i \rightarrow X$ lifts uniquely to $\tilde{X}$

Since $\sigma(\text{Interior of } \Delta^i) \subseteq X^{\text{smooth}}$ & π is isom. over X^{smooth} there is uniqueness



For $a \in \Delta^i$, $B_\epsilon(a) \cap (\Delta^i)^0$ is conn'd, so $\pi^{-1}(B_\epsilon(a) \cap (\Delta^i)^0)$ is connected. $K_\epsilon \subseteq \pi^{-1}(\sigma(\Delta^i))$: cpt (bc π is proper)

$\bigcap_{\epsilon > 0} K_\epsilon$ is non- $\emptyset$ & conn'd (bc they are nested & $\neq \emptyset$ compact sets) $\leadsto$ so $\nexists$ intersection is a pt. & can lift uniquely

not true for arbitrary resolutions

8.5

Note on functoriality of IH:

Prop. Suppose $f: Y \rightarrow X$ is a finite map. Then there are Whitney stratifications

$$Y = Y_n \supset Y_{n-1} \supset \dots \supset Y_0 \quad \& \quad X = X_n \supset X_{n-1} \supset \dots \supset X_0$$

s.t. $Y_i = f^{-1}(X_i)$.

Pf: Build stratifications together.

At each stage of induction, remove points, their preimages, & their images whenever they fail the 2 Whitney conditions. As this is a closed condition, we can then continue the induction downwards on dimension.

This has applications beyond normal varieties!

e.g. $q: Y \rightarrow Y/G$ for a finite group G acting on a normal variety Y .

Exercise: Show $IH_*^P(Y/G, \mathbb{Q}) = IH_*^P(Y, \mathbb{Q})^G$.

⑨ Ex: Back to Whitney umbrellas:

$$X = \text{Spec}(A) = \{x^2 = y^2 z\} \subseteq \mathbb{C}^3$$

$$A = \mathbb{C}[x, y, z] / (x^2 - y^2 z) \hookrightarrow \mathbb{C}[u, v]$$

$$\begin{array}{ccc} x & \mapsto & uv \\ y & \mapsto & u \\ z & \mapsto & v^2 \end{array}$$

$$A \cong \mathbb{C}[u, uv, v^2] \leftarrow \text{Can get } v \text{ from } x^2 - (y^2 z)$$

$$\bar{A} = \mathbb{C}[u, v]$$

$$\text{IH}(X; \mathbb{Z}) = \text{IH}(\mathbb{C}^3; \mathbb{Z})$$

For projective varieties, can glue normalizations for affine charts.

e.g. $X = \{x^2 w = y^2 z\} \subseteq \mathbb{P}^3$

On $w \neq 0$: $v_w: \mathbb{A}_{u,v}^2 \rightarrow X$
 $(u, v) \mapsto [uv : u : v^2 : 1]$

On $z \neq 0$: $v_z: \mathbb{A}_{a,b}^2 \rightarrow X$
 $(a, b) \mapsto [a : ab : 1 : b^2]$

glued where $v \neq 0$ / $b \neq 0$ with

$$\left[\frac{u}{v} : \frac{u}{v^2} : 1 : \frac{1}{v^2} \right] = [a : ab : 1 : b^2]$$

$$\boxed{a = \frac{u}{v}}, \boxed{b = \frac{1}{v}}$$

Tot($\mathcal{O}_{\mathbb{P}^1}(1)$) $\xrightarrow{\mathbb{P}^1 \text{ gluing}}$

+ $w = z = 0$ compactifies Tot($\mathcal{O}_{\mathbb{P}^1}(1)$)

$$\Rightarrow \bar{X} = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(1)) \quad \text{"Hirzebruch surface"}$$

$$\text{IH}_i(X, \mathbb{Q}) = H_i(\bar{X}, \mathbb{Q}) \xrightarrow{\uparrow} H_i(\mathbb{P}^1, \mathbb{Q}) \oplus H_{i-2}(\mathbb{P}^1, \mathbb{Q})$$

projective bundle formula
[See back]

$$= \begin{cases} \mathbb{Q}, & i=0 \\ 0, & i=1 \\ \mathbb{Q} \oplus \mathbb{Q}, & i=2 \\ 0, & i=3 \\ \mathbb{Q}, & i=4 \end{cases}$$

Projective Bundle formula: $E \rightarrow X$: vector bundle.

$$H^*(P(E); \mathbb{Z}) \cong \frac{H^*(X; \mathbb{Z})[\xi]}{(\xi^{r+1} + c_1(E)\xi^r + c_2(E)\xi^{r-1} + \dots + c_{r+1}(E))}$$

As $H^*(X; \mathbb{Z})$ - algebras

Cor: $H^k(P(E); \mathbb{Z}) \cong \bigoplus_{i=0}^r H^{k-2i}(X; \mathbb{Z}) \quad \forall k.$

⑩ Rmk: IH is not a homotopy invariant!

e.g. $S^4 \sqcup \mathbb{CP}^2$

normal $\searrow$
 $S^4 \cup_{S^2} \mathbb{CP}^2 \sim_{\text{homotopy equiv.}} S^4 \vee S^4$

$S^4 \sqcup S^4$

normal $\swarrow$

$\Rightarrow IH_i^p(S^4 \vee S^4, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(S^4, \mathbb{Z}) \leftarrow \text{different!}$
 $IH_i^p(S^4 \cup_{S^2} \mathbb{CP}^2, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(\mathbb{CP}^2, \mathbb{Z})$

Note: In both formulas above, answer is independent of p ! This reflects smoothness of normalization, as we prove below:

Thm. X : oriented, normal, cpt n -diml pseudomfd / X : projective, normal (real dim n) complex variety.

Then: $IH_i^t(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$
 $IH_i^0(X, \mathbb{Z}) \cong H^{n-i}(X, \mathbb{Z})$

$\begin{cases} 0 = 0\text{-perversity fn} \\ t = \text{top perversity fn} \\ t(i) = i-2, i \geq 2. \end{cases}$

[Rmk. Other perversities "interpolate" between H_i & H^{n-i} .]

Pf Ideas: LemA: $IH_0^p(Y) = H_0(Y^{sm})$

[Compute w/ chains: $\dim(\{ \xi \} \cap Y_{n-2}) \leq -2$ for $\xi \in IC_0^p$
 $\& \dim(\{ \xi \} \cap Y_{n-2}) \leq -1$ for $\xi \in IC_1^p$]

$\Rightarrow IH_0^p$ same as computing H_0 for $Y^{sm} = Y - Y_{n-2}$.

LemB L : compact of dim. $k-1$, with Whitney stratⁿ $L = L_{k-1} \supset L_{k-2} \supset \dots \supset L_0$

$IH_i^p(\text{cone}(L)) = \begin{cases} IH_i^p(L) & , i < k-1-p_k \\ 0 & , i \geq k-1-p_k \end{cases}$

Pf: $\text{cone}(L)$ has stratⁿ $\text{cone}(L)_i = \begin{cases} \text{cone}(L_{i-1}) & , i \geq 1 \\ \{v\} & , i = 0 \end{cases}$

cone point

$\xi \in IC_i^p(\text{cone}(L))$ is allowable at v iff:

$0 \leq i - k + p_k \iff i \geq k - p_k$

If $i \leq k - p_k - 1$, $\xi \in IC_i^p(\text{cone}(L))$ can't touch v &

$IC_i^p(\text{cone}(L)) \cong IC_i^p(L \times (0,1)) \cong IC_i^p(L) \Rightarrow \begin{cases} \text{for } i < k-1-p_k \\ IH_i^p(\text{cone}(L)) = \\ IH_i^p(L) \end{cases}$

MOVE TO LATER

⑪ If $i \geq k - p_k - 1$, form $c\xi = \text{cone}(\xi)$ for $\xi \in IC_i^P(\text{cone } L)$

By hypothesis, $\dim(|c\xi| \cap \{v\}) = 0 \leq (i+1) - k + p_k$

$$\& \dim(|c\xi| \cap \text{cone}(L_{k-1-j})) \leq \dim(|\xi| \cap L_{k-1-j}) + 1$$

So $c\xi \in IC_{i+1}^P(\text{cone } L)$. $\leq i - j + p_j + 1$

Easy to check: $\partial(c\xi) = \xi \Rightarrow IH_i^P(X) = 0$ for $i \geq k - p_k - 1$

Lem: $IH_*^P(\mathbb{R}^m \times Y) \cong IH_*^P(Y)$

[Pass to normalizations & split as product. Use $H_*(\mathbb{R}) \cong \mathbb{R}$ in deg.]

Mayer-Vietoris for IH: For $U, V \subseteq X$ open, we have a seq of complexes:

$$0 \rightarrow IC_*^P(U \cap V) \xrightarrow{\xi \mapsto (\xi, \xi)} IC_*^P(U) \oplus IC_*^P(V) \xrightarrow{(\alpha, \beta) \mapsto \alpha - \beta} IC_*^P(U \cup V) \rightarrow C$$

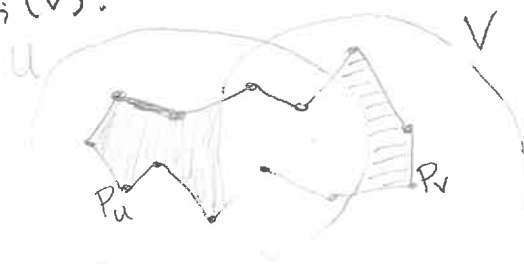
[so we have a les.

$$\dots \rightarrow IH_i^P(U \cap V) \rightarrow IH_i^P(U) \oplus IH_i^P(V) \rightarrow IH_i^P(U \cup V) \xrightarrow{\partial} IH_{i-1}^P(U \cup V)$$

Pf: Most is formal from usual MV sequence.

Hardest Part: If $\xi \in IC_i^P(U \cup V)$, then $\xi = \xi_u + \xi_v$ for $\xi_u \in IC_i^P(U)$ & $\xi_v \in IC_i^P(V)$.

Pf Sketch



define $P = |\xi|$

$$f: P \rightarrow \mathbb{R} \text{ cts. st. } f(v) = \begin{cases} 0 & \text{if } v \in \{\text{closed simplex intersecting } P_u\} \\ 1 & \text{if } v \in \{\text{closed simplex intersecting } P_v\} \\ 1/2 & \text{else} \end{cases}$$

Shrink simplices enough s.t. no simplex intersects both P_u & P_v . AND $|\xi| \cap X_i$ are sub-complexes

Pick $c \in (0, 1)$ avoiding vertices, & $\xi_u = \text{part of } \xi \text{ w/ } f < c$ & $\xi_v = \text{part of } \xi \text{ w/ } f > c$.

Compatibility is clear for $|\xi_u|$ & $|\xi_v|$.

$$|\partial \xi_u| \subseteq |\partial \xi| \cup f^{-1}(c)$$

"Thin Wall Lemma": For every subpolyhedron $Y \subseteq P$, $\dim(f^{-1}(c) \cap Y) \leq \dim(Y) - 1$

⑫ Pf of IH^0 & $IH^{\pm}$ for normal X :

Claim: If $F \& G : \{ \text{open sets in } X \} \longrightarrow \{ \text{graded abelian grps} \}$ both satisfy:

- ① $F(U) \longrightarrow F(V)$ for $U \hookrightarrow V$
- ② $F \& G$ satisfy M.V. exact sequences
- ③ $F \& G$ commute w/ direct limits over increasing unions.

Then if $\phi: F \longrightarrow G$ is a natural transformation ^{which} is an isomorphism when $V \cong \mathbb{R}^n \subset X^{sm}$ or $V \cong \mathbb{R}^{n-k} \times \text{cone}(L)$, then ϕ_X is an isomorphism.

Pf: Use MV sequences: • if ϕ is an isom. on $U, V, U \cup V$, then $\phi_{U \cup V}$ is isomorphism.

• Induct on Whitney stratification, covering X_{n-i} at each step

Thm Part 1: $IH_i^{\pm}(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$

Pf: need to check for $\text{cone}(L)$:

i	$IH_i^{\pm}(\text{cone } L)$	$H_i(\text{cone } L)$
≥ 1	$\rightarrow 0$	0 [cone L is contractible]
0	$IH_0^{\pm}(L) = H_0(L^{sm})$	$\mathbb{Z}$

$\dim(\mathbb{R}^n \cap X^{sm}) \leq i-2$
 $\forall$ coh degs. k .
 $\Rightarrow IC_*^{\pm} \subseteq C_*^{PL}$
 $\Rightarrow IH_*^{\pm} \rightarrow H_*$

$k-1-(k-2)=1$, so follows from Lem B, then A

Agree when L^{res} is connected $\Leftarrow$ top. normality. //

Thm Part 2: $IH_i^0(X, \mathbb{Z}) \cong H^{n-i}(X; \mathbb{Z})$

$V = \mathbb{R}^{n-k} \times \text{cone}(L)$: map comes from $\sigma \mapsto D(\sigma) =$ dual block of barycentric subdivision.

i	$IH_i^0(V)$	$H_c^{n-i}(V)$
$\leq k-2$	$IH_i^0(L)$	$H^{k-1-i}(L)$
$k-1$	$\rightarrow 0$	$\xleftarrow{\text{when } L \text{ is connected, i.e. normality}} \tilde{H}^0(L)$
$\geq k$	$\rightarrow 0$	0

$k-1-0=k-1$, so Lem B computes

↑ This map requires starting with a triangulation!
 Better: use the cap product. See Back.

Prop. [Cap product factors through IH]: $\forall$ perversity p ,
there is a factorization

$$\begin{array}{ccccc}
 H^{n-k}(X; \mathbb{Z}) & \xrightarrow{\cap [X]} & H_k(X; \mathbb{Z}) & & \\
 \downarrow & & \uparrow & & \\
 IH_k^p(X; \mathbb{Z}) & \longrightarrow & IH_k^p(X; \mathbb{Z}) & \longrightarrow & IH_k^{\pm}(X; \mathbb{Z})
 \end{array}$$

- ▷ Can be proved using a triangulation of X + explicit formula for cap product.
- ▷ We will defer proof as it follows more cleanly from sheaf-theoretic constructions.

(12.5) Borel-Moore Homology & $I^{BM}H$:

(Introduce/motivate BM homology)
 - MV for BM
 - Compute for cones.

X : smooth, but non-compact, oriented mfd.

→ duality & many applications of cohomology require compact supports

ea. $H_c^i(X, \mathbb{Z}) \cong H_{n-i}(X, \mathbb{Z})$

$H^i(X, \mathbb{Z}) \cong \underline{H}_{n-i}^{BM}(X, \mathbb{Z})$

"Borel-Moore" → generalizes well to IH

Borel-Moore Homology: Replaces singular chains $\sum_{k=1}^K \alpha_k \cdot \sigma_k$ with infinite sums $\sum \alpha_\sigma \cdot \sigma$ s.t. $\forall$ compact $K \subseteq X$, only finitely many σ intersect K .

Ex: for $\mathbb{R}$, we have $\sum_{i \in \mathbb{Z}} [i, i+1] \in C_1((\mathbb{R}))$

~~can~~ Can use a triangulation: $\mathbb{R} = \bigcup_{i \in \mathbb{Z}} [i, i+1]$.

$$\begin{array}{ccccc} 0 & \longrightarrow & \prod_{n \in \mathbb{Z}} \mathbb{Z} & \longrightarrow & \prod_{n \in \mathbb{Z}} \mathbb{Z} \longrightarrow 0 \\ & & (a_n) & \longmapsto & (a_{n+1} - a_n) \\ & & H_1 = \mathbb{Z} \cdot (1)_{n \in \mathbb{Z}} & & H_0 = 0 \end{array}$$

Fact: If $\bar{X}$ is cpts of X &
 $D = \bar{X} - X$,
 $H_c^i(X, \mathbb{Z}) \cong H^i(\bar{X}, D; \mathbb{Z})$

If M is compact, it is clear (from chains) that $H_i = H_i^{BM}$.

Can do the same modification to get $I^{BM}H_i^p$. All results we discussed for IH hold for $I^{BM}H$ (compute by normalizations, MV sequences, etc.).

Goal: $I^{BM}H_i^p(cL) = ?$ for $\dim(L) = k-1$; L : compact (w/o bdy) prebmfld.

Sublem: $I^{BM}H_i^p(X; \mathbb{Z}) \cong I^{BM}H_{i+1}^p(X \times \mathbb{R}; \mathbb{Z})$

Pf: Still splits as a product, but $I^{BM}H_i^p(\mathbb{R}, \mathbb{Z}) = \begin{cases} 0 & i=0 \\ \mathbb{Z} & i=1 \end{cases}$

Prop: $I^{BM}H_i^p(cL, \mathbb{Z}) = \begin{cases} IH_{i-1}^p(L, \mathbb{Z}) & i > k-1-p_k \\ 0 & i \leq k-1-p_k \end{cases}$ $\nearrow$ inequalities reversed from standard case!

Pf: Same condition at cone point: $\dim(\xi \cap \{0\}) \leq i - k + p_k$

so no intersection when $i \leq k-1-p_k$ [when $i - k + p_k \leq -1$]

For $\xi \in IC_i^p(L \times (0, \infty))$ form $R = \sum_{m \geq 0} [m, m+1]$ on $[0, \infty)$ & $\xi \times R$.
 (missing 0)

& consider $\eta = F_*(\xi \times R)$: composing $\xi \times R$ with $F: L \times (0, \infty) \times [0, \infty)^+ \rightarrow L \times (0, \infty)$

~~BM~~ $\partial(\xi \times R) = \partial \xi \times R + (-1)^i \xi \times \partial R \stackrel{\partial \xi = 0}{=} (-1)^{i+1} \xi \times \{0\} \Rightarrow \partial \eta = \xi$

12.75

So this "coring off to infinity" procedure shows

$$I H_i^P(cL) = 0 \text{ for } i \leq k-1-p_k.$$

For higher degrees $\xi \in IC_i^P(L) \Rightarrow c_\infty(\xi) = \overline{\xi \times R} \approx \xi \times [0, \infty)$

$$c_\infty(\xi) \in I^{BM} C_{i+1}^P(L)$$

$$c_\infty : IC_*^P(L) \longrightarrow IC_{*+1}^P(cL)$$

$$\partial(c_\infty \sigma) = -c_\infty(\partial \sigma)$$

for $*$ large enough
to go through cone pt. ϵ

//

(13) Poincaré Duality for IH :

Recall: $\pm(i) = \begin{cases} 0, & i \leq 2 \\ i-2, & i > 2 \end{cases}$ is the maximal/top perversity and

$\underline{0}(i) = 0 \forall i$ is the zero perversity.

▷ For p : perversity fcn, define $q = \pm - p$ to be the 'complementary perversity'.

Thm: X : projective complex variety; p & q = complementary.

Then there is a non-degenerate bilinear pairing

$$IH_i^p(X; \mathbb{Q}) \times IH_{n-i}^q(X, \mathbb{Q}) \xrightarrow{\cap} \mathbb{Q}.$$

Notes: ▷ similar versions when X is not projective adding suitable "cpt. supports" conditions.

▷ when X is smooth this is constructed by taking $\sigma_i: \Delta^i \rightarrow X$ & $\sigma_{n-i}: \Delta^{n-i} \rightarrow X$, nudging them until they are transverse, & sending to $\#(\sigma_i! \cap \sigma_{n-i}!)$. Proof of non-degeneracy comes from local computations + M.V. sequence

Idea for proof is similar: take chains s.t. $\sigma_i! \cap \sigma_{n-i}! \subseteq X^{\text{sm}}$ & is finitely many points. Using M.V., reduces to showing the corresponding statement for cones $\text{cone}(L)$.

Recall: for $\dim(L) = m$,

$$IH_i^p(\text{cone } L, \mathbb{Z}) = \begin{cases} IH_i^p(L; \mathbb{Z}) & , i < m - p_{m+1} \\ 0 & , \text{else} \end{cases}$$

$$I^{BM}H_i^p(\text{cone } L, \mathbb{Z}) = \begin{cases} 0 & , i \leq m - p_{m+1} \\ IH_{i-1}^p(L; \mathbb{Z}) & , \text{else} \end{cases}$$

If L satisfies P.D. (induction on $\dim$), then $\forall i < m - p_{m+1}$

$$IH_i^p(\text{cone } L, \mathbb{Q}) \cong IH_i^p(L, \mathbb{Q}) \xrightarrow{\downarrow} IH_{n-i}^q(L, \mathbb{Q})^* \cong I^{BM}H_{m+1-i}^q(\text{cone } L, \mathbb{Q})^*$$

& if $i \geq m - p_{m+1}$, since $m-1 = p_{m+1} - q_{m+1}$,

$$IH_i^p(\text{cone } L, \mathbb{Q}) = 0 = I^{BM}H_{m+1-i}^q(\text{cone } L, \mathbb{Q})$$

Later, we'll return. & reprove this with sheaves/Verdier duality.

(13.5) Remark: Here, the complementary perversities appear in local calculations (for cones). Another way to see complementary perversity fns must appear is through the heuristics of intersection #'s:

~~an~~ an allowable i -chain σ_i & $n-i$ chain σ_{n-i} must not meet on X_{n-k} . Their expected intersection dim. is

$$(i-k+p_k) + (n-i-k+q_k) - (n-k) = p_k + q_k - k.$$

To make this < 0 , need $p_k + q_k \leq k-1$.

For homologies to also miss each other: $p_k + q_k \leq k-2$.

Making this equal allows "as many cycles in dual as possible" $\longrightarrow$ makes pairing non-degenerate.

(14) Note there are 2 "middle" perversities:

$$m = [0, 0, 0, 1, 1, 2, 2, 3, 3, \dots] \text{ "upper middle"}$$

$$n = [0, 0, 0, 0, 1, 1, 2, 2, 3, 3, \dots] \text{ "lower middle"}$$

$m+n = t$ are complementary.

For general pseudomanifolds (locally conical mfd's), $IH^m \neq IH^n$
e.g. real algebraic varieties don't necessarily have $IH^m = IH^n$

Prop. If all strata in a Whitney stratification of X are even dimensional, then $IH_*^m(X) = IH_*^n(X)$.

ex: X : complex algebraic variety.

Pf: Since strata are even-dim'l, they only see $p(2i)$ for $\dim_{\mathbb{C}}(X) = \frac{1}{2} \dim_{\mathbb{R}}(X)$. //

In this case, $IH_*^m(X) = IH_*(X)$ is self-dual under Poincaré Duality.

e.g. If X : cx alg. variety has smooth normalization, then $IH_*^p(X) = IH_*(X)$ $\forall$ perversity fcn's p & so $IH_*^p(X)$ satisfy Poincaré Duality.

Introduces new invariants associated to X

e.g. If X has even complex dimension $2n$, get a non-deg. pairing $IH_{2n}^m(X, \mathbb{Q}) \otimes IH_{2n}^n(X, \mathbb{Q}) \rightarrow \mathbb{Q}$.

$$\sigma(X) := \#(\text{positive eigvals}) - \#(\text{negative eigvals})$$

"signature of X " $\rightarrow$ topological (bordism) invariant of X .

①5 Crash Course on Sheaf Theory:

New Goal: Sheafify everything we talked about.

$IH_i \rightsquigarrow IC$ sheaves

We want this for a few reasons:

- ▷ Can start to deal with families.
- ▷ Can understand relationships between IC sheaves on subvarieties of ambient spaces/sources & targets of maps
- ▷ Categorifies constructions in geometric rep. theory, number theory/Langlands program, etc.

Def. A presheaf is a contravariant functor

$$F: \left\{ \begin{array}{c} \text{open sets} \\ \text{in } X \end{array} \right\} \longrightarrow \left\{ A\text{-modules} \right\}$$

$$F(U) =: H^0(U, F) =: \Gamma(U, F) \quad \left(\begin{array}{l} \text{e.g. abelian grps.; } \mathcal{O}_X\text{-modules;} \\ \text{vector spaces; ...} \end{array} \right)$$

F is a sheaf if it also satisfies the gluing property:

• If $s_i \in \Gamma(V_i, F)$ s.t. $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j} \forall i, j$

Then $\exists! s \in \Gamma(\bigcup_i V_i, F)$ s.t. $s|_{V_i} = s_i \forall i$

[Note: might be infinitely many V_i !]

e.g. ① $M \in A\text{-mod}$.

$\underline{M}_X =$ constant sheaf on X

$$\underline{M}_X(U) := \left\{ \text{cts. } f: U \rightarrow M \right\} \cong M^{\# \pi_0(U)}$$

② "Structure sheaf" of X , $\mathcal{O}_X = \{ \text{sheaf of fns on } X \}$

If X is a scheme & $U = \text{Spec}(R) \subset X$ is an affine patch, $\mathcal{O}_X(U) = R$.

Def. The stalk of F @ $x \in X$ is $F_x = \varinjlim_{x \in U} F(U)$

①6

There are 2 flavors of sheaves that will be interested in: coherent & constructible

Coherent: Any algebraic variety, X , comes with a structure sheaf $\mathcal{O}_X$ of fns on X .

$$\mathcal{O}_X(U) = \{ \text{algebraic fns on } U \}$$

A coherent sheaf is a sheaf $\mathcal{F}$ on X s.t. $\mathcal{F}(U)$ is a fin. presd $\mathcal{O}_X(U)$ -module for all small enough U

If $U = \text{Spec}(A) \subseteq X$ is an affine (Zariski) open set, $\mathcal{F}|_U = \tilde{M}$ for a f.g. A -module M

(i.e. $\mathcal{F}(U - \{a\}) = M_{(a)}$)

Constructible:

Def. An A -local system on X is a locally constant sheaf $\mathcal{L}$ of A -mods. i.e. $\exists$ open covering $X = \bigcup U_i$ s.t. $\mathcal{L}|_{U_i} \cong \tilde{M}_i$ for an A -mod M_i .

$\triangleright$ If X is connected $M_i = M \forall i$ is constant.
 $\triangleright$ If $M = M_i = A \forall i$, then $\mathcal{L}$ is of "rank n ".

Exs: ① (from topology) $\mathcal{O}_X^{\text{naive}}(U) = H_n(X, X - U; A)$
 $\mathcal{O}_X^{\text{naive}} = (\mathcal{O}_X^{\text{naive}})^{\text{sh}}$

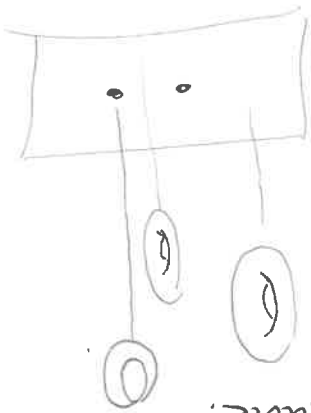
$\mathcal{O}_X$ is constant $\Leftrightarrow X$ is orientable.

② (family) $X = \{y^2 = x(x-1)(x-\lambda)\}$

f

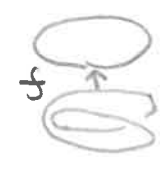
$A_1^2 \setminus \{0, 1\}$

$F(U) = H_1(f^{-1}(U), \mathbb{Z})$ stalls are $\mathbb{Z}^2$



How to tell what a locally constant sheaf is? (Or even whether it is nontrivial??)

A: Walk along paths in X & see if you end up in the same place!



Ex: Consider double cover of S^1
 $f(u) := f^* \Delta_{S^1}(u) = \Delta_{S^1}(f^{-1}(u))$

$f^* \cong A^2$ (rank 2 local system) is locally constant.

Can trivialize $S^1 = U_1 \cup U_2$

$$f|_{U_i} \cong A^2_{U_i}$$

$$(a_1, a_2) \in f(U_1) \rightarrow (a_1, a_2) \in f(U_2) \rightarrow (a_2, a_1)$$

$$\rightarrow \pi_1(S^1, x_0) \rightarrow \text{Aut}(A^2) \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

order 2 image

Prop. X : connected top. space. Then TFAE:
 (a) A - local systems on X w/ fibers M
 (b) representations $\rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M)$.

Pf: ① ρ : local system $\Rightarrow \rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M)$

Bc: $\rho: [0, 1] \rightarrow X$ gives a trivial local system
 $\rho^{-1} \rho_0 \cong \rho_0$ & $\rho^{-1} \rho_1 \cong \rho_1$.

② $\rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M) \Rightarrow \rho$: local system

Bc: ρ = sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$

for $\tilde{X} \rightarrow X$: universal cover.
 Sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$ has monodromy:
 $\rho_p = \rho$

$T_x \pi_1(X, x_0)$ acts by $T_x(s(\tilde{g} \cdot x_0)) = s(\tilde{g}(1)) = s(g \cdot x_0) = \rho(g) s(x_0)$
 $s \circ \rho^{-1}(u) \rightarrow M$ loc. const. s.t. $s(g \cdot x) = \rho(g) \cdot s(x)$

$$\tilde{\rho}_p = \tilde{\rho}: \tau: \tilde{M} \xrightarrow{\sim} p^* \mathcal{L} \text{ Deck gp. } \pi_1, \tau_{g \cdot x} = T_x \circ T_x = \tau_x \circ \rho(p)$$

18 (Harder example) Recall the elliptic curve example.

Consider $\mathcal{E}(u) = H_1(f^{-1}(u))$ fibers $\mathbb{Z}^2$ $\pi_1(A_1^2 - 30, 13) \cong F_2$.

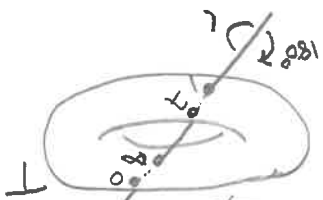
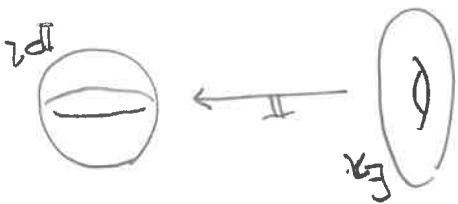
$\mathcal{Q}$: What is $p: F_2 \rightarrow \text{Aut}(\mathbb{Z}^2)$ corresponding to $\bar{A}$:

$\bar{A}: p(e_i) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ "Dehn twist"

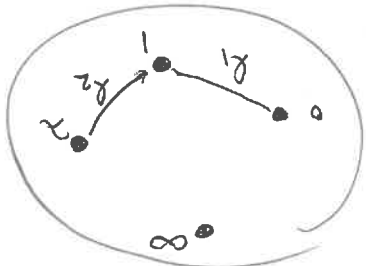
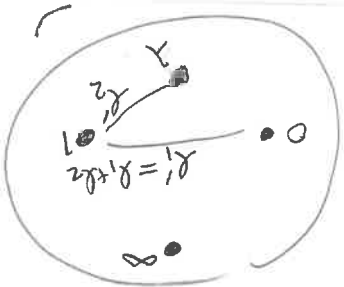
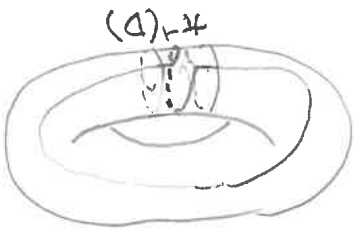


(see in Girard's notes...)

$\overline{pf}$:



$T/\mathbb{Z} \cong S^2$



In larger family $X = \{y^2 = (x+a)(x-b)(x-c)\}$

Pf. of Ehresmann's Thm: $f: X \rightarrow Y$: ^{proper} submersion of smooth mfd
 Ehresmann is local on target: suffices to ~~try~~ prove for $Y = \mathbb{R}^n$
 Submersions have local coordinates @ $x \in X$ given by
 $\mathbb{R}^{n+m} \xrightarrow{\text{proj}} \mathbb{R}^n$.

Choose partition of unity for open cover of $X \Rightarrow$ lift $\partial_i \in \mathcal{X}(\mathbb{R}^n)$
 to $v_1, \dots, v_n \in \mathcal{X}(X)$ $[v_i = \sum_{\alpha} p_{\alpha} \frac{\partial}{\partial x_i^{\alpha}}]$ s.t. $df_p(v_i) = \partial_i \forall p$.

$\varphi^i =$ flow along v_i ; $\Phi: \mathbb{R}^n \times f^{-1}(d) \rightarrow X$ by
 $\Phi(s_1, \dots, s_n, p) = \varphi_{s_n}^n \circ \dots \circ \varphi_{s_1}^1(p)$ is an isom.

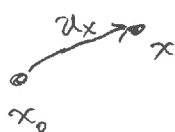
[Uses properness to show flow is well-defined/doesn't blow up & easy to check $f(\Phi(s, p)) = s$.]

Lem: $H^1(X, \mathcal{L}) \cong M_{\pi_1(X, x_0)}^{\text{fiber of } \mathcal{L} @ x_0}$ when X is conn'd

Pf: $H_0(X, \mathcal{L}) \cong M_{\pi_1(X, x_0)} = M / \langle \delta_{*} m - m \rangle$

Pf: $H_0(X, \mathcal{L}) = \left(\bigoplus_{x \in X} \mathcal{L}_x \right) / \left\langle (\alpha_* m)_{\alpha(1)} - m_{\alpha(0)} \mid \alpha = \text{path } \delta m \in \mathcal{L}_{\alpha(0)} \right\rangle$
 (1-simplex)

$\left[\phi(m_x) = u_{x,*}^{-1}(m_x) \right]$ use path to transport to,
 $M_{\pi_1(X, x_0)}$



$\triangleright \phi$ is well-defined bc. $\gamma' = u_x \circ u_x^{-1}$ is loop
 so $\gamma_* \circ u_{x,*}^{-1}(m_x) - u_{x,*}^{-1}(m_x) = 0$.
 $\parallel$
 $u_{x,*}^{-1}(m_x)$

$\triangleright$ Relations exactly match after ϕ . //

Claim: $f: X \rightarrow Y$: smooth + proper $\Rightarrow f_* \mathcal{L}$ is a local system if $\mathcal{L}$ is.

2 Main Ingredients:

① Proper base change: Suppose we have a Cartesian diagram:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

If f is proper & F is constructible, then $g^* f_* F \cong f'_* g^* F$.

Properness used for openness of fibers & "tube lemma" $\lim_{\leftarrow} F(f^{-1}V) \cong \Gamma(X_s, F|_{X_s})$

Cor:

$$\begin{array}{ccc} X & \rightarrow & X \\ \downarrow f & & \downarrow f \\ \text{Spec}(k) & \xrightarrow{s} & S \end{array}$$

Then $(F|_{X_s})(X_s) = f_* F|_s$ for $s \in S$.

i.e. fiber of $f_* F$ is $\Gamma(F|_{\text{fiber}})$.

Notes: proper is necessary! $f: \mathbb{A}^2 \setminus \{(0,0)\} \xrightarrow{\pi_2} \mathbb{A}^1$
 $f_* \mathcal{O}_{\mathbb{A}^2 \setminus \{(0,0)\}} \cong \mathcal{O}_{\mathbb{A}^1}[x]$ & $\Gamma(\mathcal{O}_{\mathbb{A}^1 \setminus \{0\}}, \mathbb{A}^1 \setminus \{0\}) = k[x, x^{-1}]$
 $\Rightarrow f_* \mathcal{O}|_0 = k[x] \leftarrow \text{not equal!}$

MORE ON BACK!

② Ehresmann's Thm: A proper submersion of smooth mfd's is a locally trivial fibration, i.e. $\forall y \in Y, \exists \text{ neighd } U \ni y \text{ s.t. } f^{-1}(U) \cong U \times f^{-1}(y)$.

~~By ②, $f^{-1}(U) \cong U \times f^{-1}(y)$ is locally trivial & $\mathcal{L}$ is locally trivial.~~

Δ Can argue claim only with ② actually! ① is just needed to say $f_* \mathcal{L}$ measures $H^i(X_y, \mathcal{L}|_{X_y})$.

Prop: $f: X \rightarrow Y$: proper map & $\mathcal{L}$: local system (or constructible sheaf) on X .

Then $f_* \mathcal{L}$ is constructible on Y .

Pf: uses a powerful result on Whitney stratifns:

* Thom's first isotopy lemma: If $X = \bigsqcup_{\lambda} X_{\lambda}$ is Whitney stratifn s.t. $X_{\lambda} \rightarrow Y$ is smooth, then $f: X \rightarrow Y$ is a stratified-locally-trivial fibration. } have non-smooth X

* Existence: Build a Whitney stratifn of X & Y s.t.:

$\Delta f^{-1}(Y_{\lambda})$ is a union of strata of X

$\Delta f_{\lambda}: X_{\lambda} \rightarrow Y_{\lambda}$ is a submersion/smooth when it is defined.

$\Delta (\bigsqcup_{\lambda} X_{\lambda})$ has $\mathcal{L}|_{X_{\lambda}}$ locally constant.

* Then $f_* \mathcal{L}|_{Y_{\lambda}}$ is a local system by isotopy lemma $\Rightarrow f_* \mathcal{L}$ is constructible //

(Co)Homology with coefficients in a local system:

for $\sigma: \Delta^i \rightarrow X$ & $\mathcal{L}$: local system $\Rightarrow \sigma^* \mathcal{L}$: loc. sys. on Δ^i
 Note Δ^i = contractible, so $\sigma^* \mathcal{L}$ = constant shf.

$\mathcal{L}_\sigma := \sigma^* \mathcal{L}(\Delta^i)$ = fiber of $\mathcal{L}|_{\sigma(\Delta^i)}$; $\forall p \in \Delta^i$, get restⁿ map
 $p \in \tau \subset \sigma: P_\tau^\sigma = (P_\tau^\tau)^{-1} \circ P_p^\sigma: \mathcal{L}_\sigma \rightarrow \mathcal{L}_\tau$ $P_p^\sigma: \mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{L}_{\sigma(p)} \cong (\sigma^* \mathcal{L})_p$
 $C_i(X, \mathcal{L}) = \left\{ \sum_\sigma \mathcal{L}_\sigma \cdot \sigma \mid \Sigma \text{ is finite (modify for BM)} \right\}$

$$\partial_i \left(\sum_\sigma \mathcal{L}_\sigma \cdot \sigma \right) = \sum_\sigma \sum_{\tau: \text{face of } \sigma} \pm P_\tau^\sigma(\mathcal{L}_\sigma) \cdot \tau$$

only difference from normal homology.

$$H_i(X, \mathcal{L}) = \text{Ker}(\partial_i) / \text{Im}(\partial_{i+1})$$

Ex: $H_i(\mathbb{C}^x, \mathcal{L}_m) = ?$ [$\mathcal{L}_m$ = local system for monodromy $z \mapsto z^m$]
 e.g. $\mathbb{Z} \rightarrow \text{Aut}(A) = A^x$
 $n \mapsto \lambda^n$

$\mathbb{C}^x \sim S^1$, so suffices to do S^1 : $S^1 = \bigcirc$ has triangulation T w/ $\sigma: \Delta^1 \rightarrow S^1$

$$C_1^T(\mathbb{C}^x, \mathcal{L}_m) = C_0^T(\mathbb{C}^x, \mathcal{L}_m) = A.$$

$$0 \rightarrow A \xrightarrow{\lambda - \text{Id}_A} A \rightarrow 0$$

the isomorphism matters!

$$H_d(\mathbb{C}^x, \mathcal{L}_m) = \text{Ker}(\partial_{m-1}) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

$$H_1(\mathbb{C}^x, \mathcal{L}_m) = k / (\partial_{m-1}) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

Can define similarly for IH_i

Have all similar formulas as H_i : MV seqs, [Poincaré duality], if fibers are vs.: $\text{IH}_i(X, \mathcal{L}) \cong \text{IH}^i(X, \mathcal{L})^*$

$C_*(X, \mathcal{L}) = C_*(X_H) \otimes_{G/H} M$ & $C^*(X, \mathcal{L}) = \text{Hom}_{G/H}^*(C_*(X_H), M)$.
 $\text{HCKer}(P_2) \xrightarrow{\uparrow} \text{fiber}$ $\Delta H^0(X; \mathcal{L}) \cong M^{\pi_1(X)}$

①9 local systems are clsd under $\oplus, \otimes, \underline{\text{Hom}}, (-)^V = \underline{\text{Hom}}(\mathbb{Z}^V, \mathbb{A}_X)$

In fact, they are even better:

Prop: $\text{Loc}(X) \subset \text{Sh}(X)$ is a full abelian subcat.

Pf: $\mathcal{L} \xrightarrow{\phi} \mathcal{L}'$ a map of local systems.

$$\text{Ker}(\phi)(U) := \text{Ker}(\mathcal{L}(U) \xrightarrow{\phi(U)} \mathcal{L}'(U)) \leftarrow \text{no sheafification}$$

locally constant.

$$\text{Coker}(\phi)(U) := \text{Coker}[\mathcal{L}(U) \rightarrow \mathcal{L}'(U)]^{\text{Sh}}$$

Sheafification

Note: f_* is same as coherent case, but f^* is not!
 $f^*F = f^{-1}F$
 $f^{-1}F|_V = M|_V$ for $V = f^{-1}(U)$ s.t. $F|_U = M$

Note: NOT true for vector bundles for example:

On $\mathbb{P}^1$, have $0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_p \rightarrow 0$.

Prop. $\text{Loc}(X)$ is stable under f^* & f_* for finite covering maps.

WARNING: Not stable under f_* for general proper maps!

(fibers can jump for example!)

Constructible Sheaves: These will capture $Rf_* \mathbb{K}_X \forall f: X \rightarrow Y$ (proper)

Def. $F \in (\mathbb{K}_X\text{-mod})$ is constructible if $X = \coprod_{\alpha} X_{\alpha}$ is stratified so that $F|_{X_{\alpha}}$ is a local system $\forall \alpha$.

Ex: $f: \mathbb{C} \rightarrow \mathbb{C}$ by $z \mapsto z^m$

$f_* \mathbb{K}_{\mathbb{C}} = ?$ For $D = \text{disk not thru } 0$,

$$f_* \mathbb{K}_{\mathbb{C}}(D) = \mathbb{K}_e(f^{-1}(D)) \cong \mathbb{K}^m$$

For $D = \text{disk thru } 0$, $f_* \mathbb{K}_{\mathbb{C}}(D) \cong \mathbb{K}$.

$$f_* \mathbb{K}_{\mathbb{C}}|_{\mathbb{C}^*} \cong \mathcal{L}_{(z \mapsto z^m \text{ on } \mathbb{C}^*)} \quad \& \quad f_* \mathbb{K}_{\mathbb{C}}|_0 \cong \mathbb{K}$$

$$\begin{array}{c} \bigcirc \\ \text{D} \end{array} \Rightarrow \mathbb{K} = f_* \mathbb{K}_{\mathbb{C}}(D) \xrightarrow{\text{res.}} f_* \mathbb{K}_{\mathbb{C}}(D') = \mathbb{K}^{\oplus m}$$

is invts. under monodromy rep.

(20)

This example is more general than it seems!

If $f: C \rightarrow C'$ is a non-constant map b/w smooth curves, locally it is just $z \mapsto z^{e_x}$ for $e_x = \text{ramification index @ } x$.

[Ramification pts. are $x \in C'$ where f is not smooth!]

Detour: smooth maps in algebraic geometry.

"smoothness of a map $f: X \rightarrow Y$ " $\iff$ "submersion of (pseudo)manifolds"

Def: $\text{Spec}(A[x_1, \dots, x_n]/(g_1, \dots, g_r)) \rightarrow \text{Spec}(A)$ is smooth if $\left(\frac{\partial g_i}{\partial x_j}\right)_{\substack{1 \leq i \leq r \\ 1 \leq j \leq n}}$ is of rank r $\forall x$.

i.e. $f: X \rightarrow Y$ is smooth iff it induces surjections on tangent spaces.

non-ex: $\pi: \text{Spec } k[u, v] \rightarrow \text{Spec } k[x, y]$
by $x \mapsto u, y \mapsto uv$.

$$k[u, v] \cong k[x, y][v]/(xv - y)$$

$[x]$ vanishes @ $x=0$.

Thm: $\varphi: X \rightarrow Y$ is smooth iff φ is flat, locally of finite presentation and its fibers are smooth.

E.g. • $\varphi: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ by $z \mapsto z^m$

Fiber @ 0 is $k[x]/(x^m) \leftarrow \text{non-reduced, so not smooth!}$

- Elliptic curve ex. is smooth on $\mathbb{A}^1 - \{0, 1\}$ but not if we extend to 0 or 1

Recap on abelian categories:

Working Def. An abelian category $\mathcal{A}$ is an additive category
(i.e. s.t. $\text{Hom}_{\mathcal{A}}(A, B)$ is an abelian gp + mild conditions.) s.t.

$\forall A \xrightarrow{\phi} B$ in $\mathcal{A}$, $\exists \text{Ker}(\phi), \text{Coker}(\phi) \in \mathcal{A}$ &

$\text{Coim}(\phi) \rightarrow \text{Im}(\phi)$ is an isomorphism.
(quotient of A) (sub of B)

Defined via universal property: e.g. $\text{Coker}(\phi) \in \mathcal{A}$ is

$$A \xrightarrow{\phi} B \longrightarrow \text{Coker}(\phi)$$

$$\begin{array}{ccc} & \searrow \omega & \downarrow \exists! \\ \forall f & \searrow & \downarrow \\ \text{s.t. } f \circ \phi = 0 & & \end{array}$$

Exs: $\triangleright \text{Mod}_A = \text{cat. of } A\text{-modules}$
 $\triangleright \text{Sh}(X) = \text{cat. of sheaves on } X$

This condition arises when one is not careful w/ decorations
one adds: e.g. $\{\text{filtered vector spaces}/k\}$ is not abelian, but has
Kernels & Cokernels.

Problem: $F^p k := \begin{cases} k & \text{for } p \geq 1 \\ 0 & \text{for } p < 1 \end{cases}$ & $\tilde{F}^p k = \begin{cases} k & \text{for } p \geq 0 \\ 0 & \text{for } p < 0 \end{cases}$

$f: k \rightarrow k$ is filtered, i.e. $f(F^p k) \subseteq F^p k$.

$\text{coim}(f) = k$ w/ F -filtⁿ

$\text{im}(f) = k$ w/ $\tilde{F}$ -filtⁿ

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Sources of Constructible Sheaves: Cohomology of proper maps!

Next Goal: develop some homological alg. to define $R^i f_* \mathcal{L}$; $H^i(X, \mathcal{L})$.

Reminders on sheaf cohomology:

- Example of a derived functor: measures discrepancy of a functor from being exact.

between abelian cats

Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a functor. F is right exact (resp. left exact) if whenever $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ is s.e.s., we have an exact seq $F(A_1) \rightarrow F(A_2) \rightarrow F(A_3) \rightarrow 0$ (resp. $0 \rightarrow F(A_1) \rightarrow F(A_2) \rightarrow F(A_3)$)

Exs: f_* is left exact; $\Gamma(X, -)$ is left exact
 $- \otimes F$ is right exact.

Thm/Def: $F: \mathcal{A} \rightarrow \mathcal{B}$ a left exact functor. Then its right derived functors $R^i F: \mathcal{A} \rightarrow \mathcal{B}$ are functors s.t.:

- $\triangleright R^i F$ is additive & $= 0$ for $i < 0$.
- $\triangleright R^0 F = F$
- $\triangleright$ Have a long $\forall$ ses.
- $\triangleright$ functorial as map on ses
- $\triangleright R^i F$ is universal with these properties
- $\triangleright \forall A \in \mathcal{A}, A \hookrightarrow I$ s.t. $R^i F(I) = 0 \forall i > 0$

Some examples: $\triangleright \Gamma(X, -)$ has right derived functor $H^i(X, -)$

$\triangleright$ Derived functors of $f_*(-)$ gives $R^i f_*(-)$: sheaves whose fibers are coh. of fibers (proper base change!)

★ Thm/Def does not show how to compute! ★

Practical Guide: ~~replace~~ To compute $F(A)$, replace A by a complex $A \rightarrow C^\bullet$.

For all of our purposes, we want A & $C^\bullet$ to be "the same"

$\triangleright$ In particular, need $H^i(A) = H^i(C^\bullet)$, but this is not enough!

$\triangleright$ Rather than give formulas, let's first explore what we mean by 2 complexes being "the same".

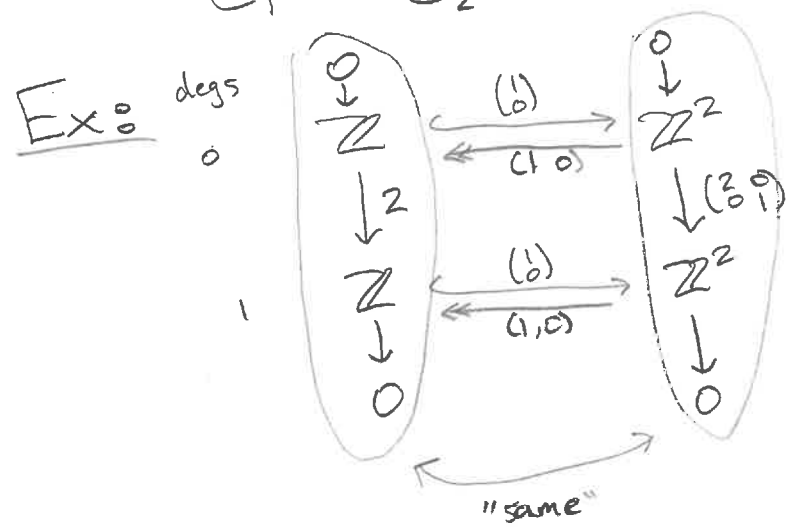
The Derived Category:

~~Abelian~~ A : abelian category (e.g. $\text{Loc}(X)$, $\text{Sh}(X)$, etc.)

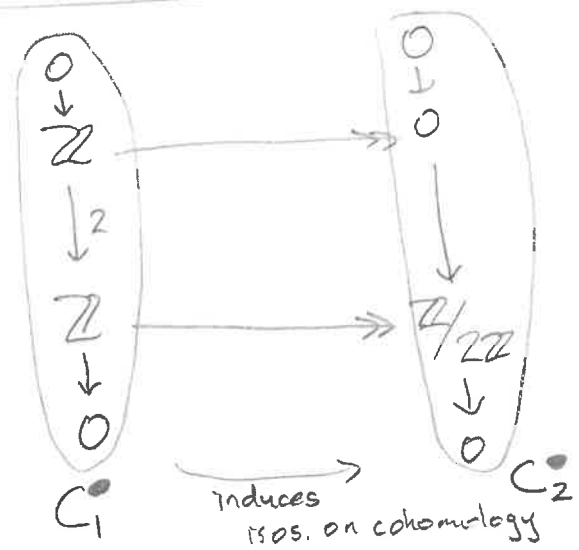
$C(A)$ = category of complexes

$$\dots \rightarrow A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow \dots \text{ in } A$$

IDEA: want to treat 2 complexes as "the same" if $C_1 \rightarrow C_2$ induces isomorphisms on (co)homology groups



call these "quasi-isomorphisms"



But: there's no inverse!

Working Def: $D(A)$ is a category whose objects are complexes & maps are usual maps + inverses of all quasi-isos.
e.g. $C_1 \cong C_2$ in $D(A)$.

Cost of this construction is that $D(A)$ is no longer abelian!

$D(A)$ is "triangulated": suppose $0 \rightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \rightarrow 0$ is a short exact sequence. For a complex $C^\bullet$, we have a "shift functor" $C^\bullet[1] = C^{\bullet+1}$

$$C^\bullet: \dots \rightarrow C^1 \xrightarrow{d^1} C^2 \xrightarrow{d^2} C^3 \rightarrow \dots$$

$$C^\bullet[1]: \dots \rightarrow C^2 \xrightarrow{d^2} C^3 \xrightarrow{d^3} C^4 \rightarrow \dots$$

$$\implies H^i(C^\bullet[1]) = H^{i+1}(C^\bullet)$$

Claim/Lem: There is a natural map $C^\bullet \rightarrow A^\bullet[1]$ in the derived category.

Pf: Put $\text{cone}(f)^\bullet: \dots \rightarrow \bigoplus_{B^n}^{A^{n+1}} \xrightarrow{\partial} \bigoplus_{B^{n+1}}^{A^{n+2}} \rightarrow \dots$

$$\partial(a, b) = (-\partial_A a, \partial_B b - f(a))$$

$$\partial = \begin{pmatrix} -\partial_A & 0 \\ f & \partial_B \end{pmatrix}$$

$$A^\bullet[1] \xleftarrow[\pi]{\text{proj}^n} \text{cone}(f)^\bullet \xrightarrow{\psi} C^\bullet$$

$$\psi(a, b) = g(b) \quad [\text{well-defined bc } \psi(\partial(a, b)) = g(\partial_B b - f(a)) = \partial_C g(b) \quad \underbrace{g \circ f = 0}]$$

ψ is a g/iso:

$$\text{Ker}(\psi) = \{(a, b) \mid g(b) = 0\} = A^{n+1} \oplus f(A^n) \cong A^{n+1} \oplus A^n$$

$$\implies \text{Cone}(1_A)^\bullet: \left[\dots \rightarrow \bigoplus_{A^n}^{A^{n+1}} \xrightarrow{\begin{pmatrix} -\partial_A & 0 \\ -1_A & \partial_A \end{pmatrix}} \bigoplus_{A^{n+1}}^{A^{n+2}} \rightarrow \dots \right]$$

has trivial cohomology groups
bc 1_A homotopic to 0: define

$$A^{n+1} \oplus A^n \xrightarrow{s} A^n \oplus A^{n-1}$$

$$s(a, a') = (-a', 0)$$

Can check $ds + sd = 1_A$.

Reminder on homotopies:

Lem. If $f, g: A^\bullet \rightarrow B^\bullet$ are homotopic, i.e.

$$\begin{array}{ccccccc} \dots & \longrightarrow & A^{n-1} & \xrightarrow{\partial_A} & A^n & \xrightarrow{\partial_A} & A^{n+1} \longrightarrow \dots \\ & & \downarrow f & \swarrow s & \downarrow f & \swarrow s & \downarrow f \\ \dots & \longrightarrow & B^{n-1} & \xrightarrow{\partial_B} & B^n & \xrightarrow{\partial_B} & B^{n+1} \longrightarrow \dots \end{array} \quad \text{s.t.} \quad f - g = \partial_B s + s \partial_A$$

Then f & g induce the same map on cohomology groups

Pf: $f(x) - g(x) = \partial_B(s(x)) + s(\partial_A x) = \partial_B(s(x))$ is a bdrly
 $\uparrow x \in \text{Ker}(\partial_A) \Rightarrow [f(x)] = [g(x)] \text{ in } H^n(B^\bullet) //$

"Triangulated" Structure on $D^b(A) :=$ has a collection of "distinguished triangles"

① Every map $A^\bullet \xrightarrow{f} B^\bullet$ fits into a unique distinguished triangle

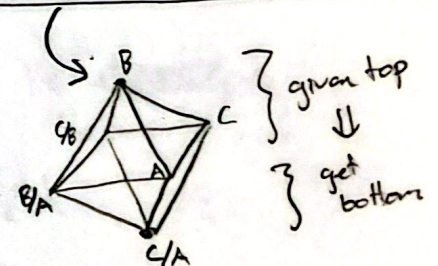
$$\begin{array}{ccc} A^\bullet & \xrightarrow{f} & B^\bullet \\ \uparrow \text{Eq} & & \downarrow \text{Cone}(f) \\ \text{Cone}(f)^\bullet & & \end{array} \quad \& \quad \text{Cone}(1_A) = 0$$

② $\{\text{distinguished triangles}\}$ is closed under $[1]$ & rotation $\left(\begin{array}{c} A^\bullet \rightarrow B^\bullet \\ \uparrow \text{Eq} \downarrow \text{Cone}(f) \end{array} \rightsquigarrow \begin{array}{c} B^\bullet \rightarrow C^\bullet \\ \uparrow \text{Eq} \downarrow \text{Cone}(g) \end{array} \right)$

③
$$\begin{array}{ccccc} A^\bullet & \rightarrow & B^\bullet & \rightarrow & C^\bullet \xrightarrow{f} \\ \downarrow & & \downarrow & & \downarrow \\ A'^\bullet & \rightarrow & B'^\bullet & \rightarrow & C'^\bullet \xrightarrow{f'} \end{array}$$

Reflection of 3rd isom. thm:
 $\exists C/A$ fitting into
 $0 \rightarrow B/A \rightarrow C/A \rightarrow C/B \rightarrow 0$

④ [Technical] Octahedral Axiom: $\forall A \rightarrow B \rightarrow B/A$
 $\& \forall B \rightarrow C \rightarrow C/B$



Notes: Hom spaces between cxs. remember
 $\star \quad \text{Hom}^n(A^\bullet, B^\bullet) = \prod_{p \in \mathbb{Z}} \text{Hom}(A^p, B^{p+n})$

$\partial^n = \prod_p \partial_p^n$ for $\partial_p^n: \text{Hom}(A^p, B^{p+n}) \rightarrow \text{Hom}(A^p, B^{p+n+1})$
 by $\partial_p^n(f_p) = \partial_{B^\bullet}^{p+n} \circ f_p \circ (-1)^{n+1} f_{p+1} \circ \partial_{A^\bullet}^p$

Lem: $\Gamma(X, \text{Hom}^\bullet(F^\bullet, G^\bullet)) = [F^\bullet, G^\bullet] = \{\text{homotopies } F^\bullet \rightarrow G^\bullet\}$
 (see Back!)

Pf that $H^0 \Gamma(X, \text{Hom}^\bullet(A^\bullet, B^\bullet)) = [A^\bullet, B^\bullet] = 0$

$$\Gamma(X, \text{Hom}^\bullet(A^\bullet, B^\bullet)) = \prod_p \text{Hom}_{O_X}(A^p, B^{p+n})$$

$$\text{Ker}(\partial^0) = \left\{ (f_p) \in \prod_p \text{Hom}_{O_X}(A^p, B^p) \mid \overbrace{\partial_B \circ f - (-1)^n f \circ \partial_A}^{\partial(f_p)} = 0 \right\}$$

↑
chain map cond

$$\text{Im}(\partial^{-1}) = \left\{ \sum \partial_B^{p-1} \circ h^{p-1} \circ \partial_A^p \mid h_p: A^p \rightarrow B^{p-1} \right\} = \{ \text{homotopies} \}$$

W

Pf. octahedral identity holds?

- Back to derived functors: -

Recall that we want to build "derived functors" $Rf_*, H^*, - \otimes^L F, \dots$

By: ① Replace F w/ a "good" complex $C^\bullet$ qiso to F .

② Compute $F(C^\bullet)$ & take cohomology.

Idea: replace F w/ qiso. complex $C^\bullet$ s.t. $R^i F(C^\bullet) = 0 \forall i > 0$.

$\rightarrow$ this depends on F !

ie. To compute $F(F)$ suffices to take a resolution of F by objects that are F -acyclic, e.g. C is F -acyclic iff any surjⁿ $B \twoheadrightarrow C \implies F(B) \twoheadrightarrow F(C)$.

Most common " F -acyclics" are projective & injective sheaves:

Projective:

$$\forall \begin{array}{c} P \\ \swarrow \downarrow \\ A \rightarrow B \rightarrow 0 \end{array}$$

Injective:

$$\begin{array}{ccccc} 0 & \rightarrow & A & \hookrightarrow & B \\ & & \downarrow & \swarrow & \\ & & I & & \end{array}$$

Warning: $\triangleright \text{Sh}(X)$ has enough injectives, but not enough

projectives

$\triangleright$ Injectives are awful: they never compute anything...

$$F \hookrightarrow \prod_{x \in X} i_{x,*} I_x \quad \text{for } I_x: \text{injective module w/ } I_x \supset F_x.$$

Injective modules: $\triangleright$ over $\mathbb{Z}$, must be divisible: $\mathbb{Q}$
(true for any PID)

$\triangleright$ If R is a domain & not a field, every injective R -module is ∞ -divisible.

Warning: "Projective" sheaf $\neq$ locally free !

locally a projective module.

Why? $F = \mathcal{O}$, $0 \rightarrow \mathcal{O}(-2) \rightarrow \mathcal{O}(-1)^{\oplus 2} \rightarrow \mathcal{O} \rightarrow 0$

Γ is not exact on this:

les: $0 \rightarrow 0 \rightarrow 0 \rightarrow k \rightarrow k^2 \rightarrow k \rightarrow 0$

What sheaves are acyclic?

▷ Injectives are acyclic $\forall$ left exact F .
 ↳ standard defn/construction uses these.

★ Flasque/flabby: when restriction maps are always surjective

↳ Prop. F : flasque $\Rightarrow F$ is Γ & f_* acyclic.

Pf. $0 \rightarrow F \xrightarrow{\alpha} G \xrightarrow{\beta} H \rightarrow 0$ s.e.s. Then $G \rightarrow H$ is

locally surjective $\Rightarrow \exists g_i \in \Gamma(U_i, G) \mapsto h|_{U_i}$

on $U_i \cap U_j, \beta(g_i - g_j) = \beta(g_i) - \beta(g_j) = h|_{U_i \cap U_j} - h|_{U_i \cap U_j} = 0 \Rightarrow \exists s_{ij} \in \Gamma(U_i \cap U_j, F)$

Extend $\tilde{f}_{ij} \in \Gamma(U_j, F)$ lift & $g_j \mapsto g_j + \tilde{f}_{ij}$ s.t. $s_{ij} = g_i - g_j$

Now $g_i = g_j$ on $U_i \cap U_j$. Then continue... $\Rightarrow H^1(X, F) = 0$

▷ For left derived functors, use flat sheaves $\mathcal{O}_F \rightarrow I \rightarrow I_1 \rightarrow 0$ Then use $L_n f_*$

$\mathcal{B}^\bullet \rightarrow F$: q.iso.

NOTE: It is still very hard to compute $f_* F$ in general! [will eventually do so w/dec. thm!]

~~General Philosophy: Better to work w/complexes than cohomology.~~

▷ Consider $R^i f_* \mathcal{L}$ in more detail: What is this sheaf roughly? ($\mathcal{L}$ = local system)

① Proper base change:

$$(R^i f_* \mathcal{L})_y = H^i(X_y, \mathcal{L}|_{X_y})$$

② If $\mathcal{L} = \mathbb{Q}_X$, Then $H^i(X, \mathbb{Q}_X) \cong H^i(X, \mathbb{Q})$

(local system on fiber.)

Pf. Let $C^n(X, \mathbb{Q}) = C^n(X)$ denote singular co-chains & $C^n: [U \rightarrow C^n(U)]$. then $\mathbb{Q}_X \rightarrow C^\bullet$ is a resolution. (uses local contractability)

C^n is flabby: can always extend a cochain on U by 0.
 $\Gamma(X, C^n) = C^n(X)$, so this proves it.

③ If not, $H^i(X, \mathcal{L}) =$ cohomology w/ $\mathcal{L}$ -coeffs [give formula + example]

Rmk: $R^i f_* \mathcal{L}$ is constructible for $\mathcal{L}$ constructible on X
 & $f: X \rightarrow Y$ proper. (Uses same proof as before!)

→ Def: $D_c(X)$ [or $D_c^b(X)$] denotes the full sub-cat.
 of $D(X) = D(\text{Sh}(X))$ [or $D_c^b(X) = D^b(\text{Sh}(X))$] of $C^\bullet$
 s.t. $H^i(C^\bullet)$ is constructible $\forall i$.

assume
also all
local systems
involved
are finite
rank!

★ $\text{Constr}(X) :=$ category of constructible shvs on X .

Prop: $D^b(X) \cong D^b(\text{Constr}(X))$

Pf Idea: for $\mathcal{L} \in \text{Constr}(X)$ & $F \in \text{Sh}(X)$, s.t. $F \twoheadrightarrow \mathcal{L}$,
 $\exists \mathcal{L}' \rightarrow F$ s.t. $\mathcal{L}' \rightarrow F \rightarrow \mathcal{L}$ is surj. & $\mathcal{L}' \in \text{Constr}(X)$
 Build $\mathcal{L}'$ by lifting local gens of $\mathcal{L}$ to F

Philosophy: complexes are better than cohomology.

Exs: ① $R(F \circ G) \cong RF \circ RG$ for left exact F, G

[vs. spectral sequence $R^p F(R^q G) \Rightarrow R^{p+q}(F \circ G)$]

② Universal coeffs: $C^\bullet =$ cx computing $H^i(X, \mathbb{Z})$.

cx computing $H^i(X; A)$ is $C^\bullet \otimes_{\mathbb{Z}}^L A$

(vs. $H^i(X; A) \cong (H^i(X, \mathbb{Z}) \otimes_{\mathbb{Z}} A) \oplus \text{Tor}^1(H^{i+1}(X, \mathbb{Z}), A)$)

Consequence: Often later try to work directly in $D_c^b(X)$.

Next Goals: ▷ 6 functor formalism in $D_c^b(X)$ [& introduction to $f_!, f^!$]

▷ Verdier duality in $D_c^b(X)$

→ opens up calculus of the derived category!

f_* & f^* are adjoint functors:

Prop. ~~f~~ $f: X \rightarrow Y$. This is a canonical isom. (functorial in each input)

$$\text{Hom}(f^*F, G) \cong \text{Hom}(F, f_*G)$$

Rmk: In coherent category, this is locally:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ \mathcal{B} & \xrightarrow{f_*} & \mathcal{C} \end{array} \quad \text{Hom}_B(M \otimes_A B, N) \cong \text{Hom}_A(M, N)$$

Special Case of:

$$\text{Hom}_B(M \otimes_A P, N) \cong \text{Hom}_A(M, \text{Hom}_B(P, N))$$

Pf: $\phi \in \text{RHS}$, $F \rightarrow f_*G$.

~~Hom~~ $\forall V \supset f^*(U), f^{-1}(V) \supset U$

$$\Rightarrow F(V) \rightarrow f_*G(V) = G(f^{-1}V) \xrightarrow{\text{res}} G(U)$$

gives $\varinjlim_{V \supset f^*(U)} F(V) \rightarrow G(U)$.

In reverse: $\psi: f^*F \rightarrow G$.

$\forall V \subset Y$ open, $f(f^{-1}V) \subset V$ gives

$$F(V) \rightarrow \varinjlim_{W \supset f(f^{-1}V)} F(W) = f^*F(f^{-1}(V)) \rightarrow G(f^{-1}(V))$$

Can check these are inverse operations on stalks //

$$\parallel f_*G(V) \xrightarrow{\text{res}} G(U)$$

Consequences of an adjoint pair (f^*, f_*) :

▷ f^* is right exact; f_* left exact [true for abelian cats.]

▷ More generally, left adj. preserves colimits: cokernels, direct limits, stalk products, pullbacks, inverse lms, ...
right adj. preserves limits: kernels, products, pullbacks, inverse lms, ...

OVER $\rightarrow$

6 functor formalism on $D_c^b(X) := \text{[a framework across many different situations]}$

Def of $f_!$: For $s \in \Gamma(U, F)$, $\text{supp}(s) = \{x \in X \mid s(x) \neq 0\}$.

e.g. If $U = \text{Spec}(A)$ & $F = \tilde{M}$ is coherent, $s \in M$ has

$$\begin{aligned} \text{supp}(s) &= \{p \in \text{Spec}(A) \mid s \neq 0 \text{ in } M_p\} \\ &= \overline{\{p \mid \exists a \in A, a \neq 0, a \cdot s = 0\}} \cong \text{Spec}(A/\text{ann}_A(s)) \\ &\quad \Downarrow \\ &\quad \{p \mid s \neq 0 \text{ in } M_p\} \end{aligned}$$

$\triangleright \Gamma_c(U, F) \subseteq \Gamma(U, F)$ is set of sections with compact support.

$\triangleright \Gamma_c$ remains left exact & so has a right derived functor $R\Gamma_c$ coh. $\rightarrow H_c^i(X, F^\bullet)$ or $H_c^i(X, F)$.] when $F = \underline{A}_X$, computes $H_c^i(X, A)$.

$\triangleright f_! F$ is given by:

$$f_! F(U) = \left\{ s \in \Gamma(f^{-1}(U), F) \mid f|_{\text{supp}(s)} : \text{supp}(s) \rightarrow U \text{ is proper} \right\}$$

$f_! F \rightarrow f_* F$ isom. for proper F . supports are relatively cpt.

$\triangleright Rf_! F$ can be computed w/ "c-soft" shvs, i.e. those for which $\Gamma(X, F) \xrightarrow{\text{res}} \Gamma(K, F)$ is surjective $\forall$ compact $K \subset X$.
[Here, $\Gamma(K, F) := \varprojlim_{U \supset K} \Gamma(U, F)$]

$\triangleright f_!$ & $Rf_!$ satisfy base change w/o properness assumptions!

Thm (Base Change).

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ Y' & \xrightarrow{g'} & Y \end{array}$$

Cartesian. Then

$$\triangleright (g')^* \circ f_! = f'_! \circ g^* \text{ in } \text{Sh}(Y)$$

$$\triangleright (g')^* \circ Rf_! = Rf'_! \circ g^* \text{ in } D^b(Y)$$

e.g. $j: U \hookrightarrow X$ open
 $j_! F = \text{extension by zero}$

/ e.g. $A^2 - \{0\} \hookrightarrow A^2$
 $j_! \mathbb{Q} = \text{extn by zero} = 0$
 $i_{k*} \mathbb{Q} = \mathbb{Q}$ $j_! \mathbb{Q} = 0$

Def of $f^!$: — Very mysterious functor in general —

Hope: maybe $f_!$ has an adjoint functor $f^!$, i.e.

$$\text{Hom}(f_! F, G) \cong f_* \text{Hom}(F, f^! G)$$

NOTE: $A = (A_n)_{n \in \mathbb{Z}}$ on $\mathbb{Z} = \{\dots \circ \circ \dots\} \xrightarrow{f} \{pt\}$

$$f_* A = \prod_{n \in \mathbb{Z}} A_n, \quad f_! A = \bigoplus_{n \in \mathbb{Z}} A_n$$

Having a left adjoint $\Rightarrow$ left exact
in abelian cat $\Rightarrow$ preserves limits

$f_!$ cannot have left adjoint bc

$$\bigoplus_n \prod_k A \neq \prod_k \bigoplus_n A$$

Cannot exist in $\text{Sh}(X)$ also bc $f_!$ is not right exact.
But maybe it exists ~~in~~ in $D_c^b(X)$:

① $j: U \hookrightarrow X$: open inclusion

$$\begin{aligned} \Gamma(U, j^! G) &= \text{Hom}_U(\underline{A}_U, j^! G) = \Gamma(U, \text{Hom}(\underline{A}_U, j^! G)) \\ &= \Gamma(U, \text{Hom}(j_! \underline{A}_U, G)) = \text{Hom}(j^* j_! \underline{A}_U, j^* G) \\ &= \text{Hom}(\underline{A}_U, j^* G) = \Gamma(U, j^* G) \end{aligned}$$

$\Rightarrow j^! = j^*$ for open inclusions.

② $j: U \hookrightarrow X$; $f: X \rightarrow Y$

$$\begin{aligned} \Gamma(U, f^! G) &= \text{Hom}_U(\underline{A}_U, f^! G) = \text{Hom}_X(j_! \underline{A}_U, f^! G) \\ &= \Gamma(Y, f_* \text{Hom}(j_! \underline{A}_U, f^! G)) \\ &= \Gamma(Y, \text{Hom}(f_! j_! \underline{A}_U, G)) \\ &= \text{Hom}_Y(f_! j_! \underline{A}_U, G) \end{aligned}$$

$\Rightarrow f^! G$ should be $[U \mapsto \text{Hom}(f_! j_! \underline{A}_U, G)]$.

NOT a sheaf! $\longrightarrow$ replace ~~$\underline{A}_X$~~ with a resolⁿ
 $\underline{A}_X \longrightarrow K^\bullet$ [by c-soft sheaves]

$\Rightarrow$ Can ~~now~~ define $f^!$ on derived category!

Method for $f^! \mathcal{G}$:

① $\mathcal{G} \rightarrow \mathcal{I}^\bullet$ an injective resolution.

$\underline{A}_X \rightarrow K^\bullet$: flat c-soft resⁿ.

If $\mathcal{I}$ is injective, $f_K^! \mathcal{I}(U) = \text{Hom}_Y(f_! K_U, \mathcal{I})$

$R: U \mapsto f_!(K^\bullet|_U)$ is a "cosheaf" valued in $\mathcal{S}(Y)$

$$\S \quad f^! \mathcal{G} = \text{Hom}_{\mathcal{D}(Y)}(R, \mathcal{G})$$

Simple Cases:

- ① If $j: U \hookrightarrow X$ is open, then $j^! \cong j^*$ [exact]
- ② If $j: Z \hookrightarrow X$ is clsd, then $j^!$ is defined on sheaves $\S$
- $$j^! \mathcal{G}(U \cap Z) = \Gamma_{U \cap Z}(\mathcal{G})$$

supp contained in $U \cap Z$

③ Better defⁿ:

$$\begin{array}{ccc} & \overset{\circ}{X} & \\ \uparrow j & & \downarrow p \\ X & \xrightarrow{f} & Y \end{array} \quad \begin{array}{l} \text{cptn} \\ p = \text{proper} \end{array} \Rightarrow f_! = p_* \circ j^!$$

Prop. $f^! = j^* \circ p^!$ (reduces to p : proper.)

for projective $X \xrightarrow{f} Y$

also have $f^! = i^! \pi^!$

$i: \text{clsd} \rightarrow \mathbb{P}_Y^n$ π : projection

$$\mathbb{P}_Y^n = Y \times \mathbb{P}^n$$

$\Delta i^!$ is done from before

$\Delta \pi$ is controlled by "dualizing α " : $\pi^! \mathcal{G} \cong \mathcal{G} \boxtimes \omega_{\mathbb{P}^n} \cong A_{\mathbb{P}^n}[2n]$

④ $f: X \rightarrow Y$ smooth:

$$f^! \mathcal{F} = f^* \mathcal{F} \otimes \omega_{X/Y}[d], \quad \omega_X = p_X^! A_{\text{pt}}$$

constr. setting: $\omega_{X/Y} = \omega_{X/Y} = \omega_X \otimes_A f^*(\omega_Y)^\vee \xrightarrow[\text{Yare smooth}]{\text{if } X \text{ S}} A_X[2d]$

$$\omega_X \otimes f^* \omega_Y^\vee$$

$d = \text{rel. dim}$

étale: $f^! \mathcal{F} = f^* \mathcal{F}(d)[2d]$

Check for a local chart $U \cong \mathbb{R}^n \rightarrow \{\text{pt}\}$ using adjunction.